FinanceFirst Bank

# Data Center Disaster Recovery Simulation

**Course: GRC205 - Incident Management and Business Continuity**

Najma Abdi Omar
12-15-2025

**GRC205 Week 21 - Practical Lab Assignment**

**Data Center Disaster Recovery Simulation**

Course: **GRC205 - Incident Management and Business Continuity**

Topic: **Business Continuity and Disaster Recovery**
Assignment Type: **Practical Simulation-Based Assignment**

**Student Name:** Najma Abdi Omar

Reg. No: **2025/GRC/10331**

**Date:** 15/12/2025

# Table of Contents

Phase 1: Initial Event (Tuesday, 6:45 AM CT) ..... 2
Phase 2: Recovery Execution (Tuesday, 9:00 AM - 12:00 PM) ..... 4
Phase 3: Complications and Decisions (Tuesday, 2:00 PM) ..... 9
Phase 4: Final Recovery and Transition (Wednesday, 8:00 AM) ..... 11
Final Deliverable: Comprehensive After-Action Report ..... 13

## Assignment Overview

In this assignment we review a disaster recovery scenario at FinanceFirst bank, FinanceFirst Bank is a regional bank with $8.5 billion in assets, serving 250,000 customers across five states (Illinois, Indiana, Wisconsin, Iowa, Michigan). The bank operates 85 branch locations, employs 1,200 staff, and offers retail banking, commercial lending, wealth management, and mortgage services.

In this scenario, we review the disaster response plans and actions undertaken by the disaster recovery manager, where each phase we receive and information to review our response and plan of action for each critical scenario.

We undertake critical decision from the moment the we receive the first message to the final full data center recovery. In the below sections, divided by task, we respond based on the information given under the task requirement.

This document is the task response document, which details each step as we progress on each phase. Additional requirements have been addressed in AAR document also included in this assignment submission.

## Phase 1: Initial Event (Tuesday, 6:45 AM CT)

**Our Response - Phase 1 (Due: Within 2 hours of incident)**

**Task 1.1: Initial Assessment and Decision**

Immediate response to this scenario:

1. **Situation Assessment:** Analyze the current situation. **What is the severity of this incident?**

   Based on the initial assessment from the message and service interruption including lack of clear timelines on when the system is estimated to be restored, we classify this incident as **critical.**

   **what are the immediate business impacts**?

   Based on the assessment and potential downtimes and system unavailability, the immediate business impact include but are not limited to: Loss of data, service unavailability, reputational damage, loss of infrastructure, business disruption, mobile, internet and core banking unavailable,

   **What information do you need that you don't currently have**?

   The information provided right now does not answer the following: Do we have enough resources, is the backup site functional and tested, and can it assist in immediate business continuity?
2. **DR Plan Activation Decision:** Should you activate the full disaster recovery plan?

   **Yes**,

   **Detailed justification.** This is because we have the main site down, systems unreachable with the possibility of recovery not certain, we have immediate business needs, peak transaction time is 2 hours away, mobile and internet banking is unreachable, wire transfer deadline will be unmet considering the downtime will exceed 18 hours if the flood issue is restored we still do not know the extent of the damage so it will take more time, considering all this and the cost implication beyond customer trust, loss of a full day business will be costlier than DR plan activation decision.
3. **Initial Actions (First 30 Minutes):** List the specific actions you will take in priority order during the first 30 minutes.

**Who will you notify?**

We will Notify the incident response team to activate the DR reactivation plan, the executive team and CEO.

**What teams will you mobilize?**

Business Continuity Coordinators, Disaster recovery team and crisis management team,

**What immediate steps will you direct?**

Coordinate the communication to avoid panic, full DR activation plan, for the crisis management communication lead ensure readiness for communication, confirm all the affected systems. Full back up restoration.

4 **Communication Plan:** Draft the initial notification message you will send to:

- ✓ **Executive Leadership (CEO, CFO, CIO, COO)**

TO: CEO, CFO, COO, General Counsel

Subject: System Outage-Primary Data Center Flooding

Executive Summary:

We are responding to a flooding incident that occurred at 6:30 AM, a water main break on the street outside the building hosting our primary data center, caused catastrophic flooding in the basement. The incident is currently classified as critical. Below is a summary of the current situation and our response.

Current Status: All production systems offline, Primary data center inaccessible due to flooding and power outage, DR site in Milwaukee is operational and ready. This affects all branch transactions, mobile and internet banking as well as wire transfers, the city says the water main break will take at least 12-18 hours to repair and after that we are yet to establish the full extent of the damage. We have activated disaster recovery and full back up restoration, activated the business continuity and disaster recovery team as well as the crisis management team. We are now restoring systems and pointing all services to our DR site. We will need additional ICT staff and support in full restoration of system. We will require approval to change operations from primary to the DR site.

We will provide the next update at 7:30 am or sooner once we establish the recovery time and data restoration.

- ✓ **Federal Reserve and FDIC (regulatory notification)**

To: Regulatory Authority

Subject: Service Interruption Notification

Service outage affecting our primary data center

Pursuant of regulatory requirement, Finance First Bank is notifying you of an interruption in their services due to flooding incident that occurred as a result of a water main break on street outside the building hosting our data center. This has affected our service including branch transactions, mobile and internet banking.

Details of the Incident: Tuesday early morning flooding that affected all services potentially affecting 25000 clients. We are monitoring the extent of the damage and will communicate with the clients individually on the next steps.

Mitigation Measures: to restore services, we are activating our DR site and service restoration

Contact information: for questions regarding this notification, please contact Najma Omar, DR Mager, +1(xxx)-xxxxxxx, drmanager@financefirst.com

Sincerely,

Finance First Bank

✓ **IT Recovery Teams**

To: IT Recovery Team

Subject: Urgent- BC/DR Response Activation

The service outage due to the flooding at the primary site has led to activation of business continuity and disaster recovery. The incident which occurred this morning at 6:45 am and classified as critical has affected all services. We are now activating the DR site and will require your full corporation in restoring the service.

We have included you in a communication channel and all communication and next step of actions will be communicated there. Please be available and standby for the next steps which will be communicated shortly.

✓ **Branch Managers**

To: Branch Manager

Subject: Urgent: Service Outage

We are experiencing service interruption due to flooding that has affected our primary data center, this has affected all systems, ATMs, mobile, internet banking and wire transfers. Please handle appropriately with walk in clients and be on standby for official communication from communication teams. We will guide you in the next step.

We regret the inconvenience as our team works on service restoration. We will also notify you on possible time for recovery once we asses extent of damage.

## Phase 2: Recovery Execution (Tuesday, 9:00 AM - 12:00 PM)

**SITUATION UPDATE - 9:00 AM**

### Our Response - Phase 2 (Due: Within 4 hours of Phase 1)

**Task 2.1: Recovery Prioritization and Execution**

Documented recovery strategy and execution plan:

5 **System Recovery Prioritization:** Given the challenges, and based on criticality and our recovery time objectives, this is our order of recovery the table also documents our planned and estimated RTOs.

1. Work on core banking restoration, without which no other system will be operational, all other systems depend on CBS and also reduce pressure from the branches.
2. Online and mobile banking to reduce customer complaints and calls to call center.
3. Branch-teller system, to serve all walk in customers.
4. Wire transfers to beat the deadlines
5. ATM Networks to allow alternative channels to be restored.
6. Email system to effectively restore communication, however this is not of utmost priority.
7. Loan originating system, this is the least of the priority based on the RTO which is 24 hours and also our current objective is to restore services and manage client panic.

All system restoration and availability from data recovery center, revert back to main site after damage assessed and repairs made.

| System | Description | RTO |
|---|---|---|
| Core Banking System | Account management, transactions, balances | 4 hours |
| Online Banking Portal | Customer web access | 2 hours |
| Mobile Banking App | Customer mobile access | 2 hours |
| ATM Network | 200 ATMs across 5 states | 6 hours |
| Wire Transfer System | Domestic and international wires | 4 hours |
| Loan Origination System | Mortgage and commercial loans | 24 hours |
| Email System | Corporate email (Microsoft 365) | 8 hours |
| Branch Teller System | In-branch transactions | 4 hours |

6 **Resource Allocation:**

Our main objective now is to have operations back up, the 15 Chicago staff will be on standby to guide and point systems to the backup site, guide the branches in how to operate, and work on system resets and ensuring everyone is back online, we will also have the ICT helpdesk coordinating efforts and communication between the teams in coordination with communication team. The 10 Milwaukee staff will be responsible for hands-on restoration, they will work on ensuring data is correct, systems are back up and coordinate the mobile banking manual restoration.

| Action | ICT Helpdesk | Chicago Staff | Milwaukee Staff |
|---|---|---|---|
| Communication and bridge between the ICT Team, The Disaster Recovery Team and Business units | R | I | I |
| Point systems to the DR site | I | R | R |
| Guide branches | R | I | I |
| System resets and Ensure everyone is back online | I | R | I |
| Hands-On restoration efforts | I | I | R |
| Data Correctness | I | I | R |
| Manual System Restoration | I | I | R |

7 **Problem-Solving:** Address the specific challenges:

- ✓ **Core banking taking longer than expected: What actions will you take?**

Communicate to the branch managers and EXCOs on the delay, assign staff with technical expertise from the main site to assist with the backup restoration.

- ✓ **Online banking configuration issues: How will you resolve this?**

Reach to the vendor partner urgently to address the unavailable step ensuring the process is undertaken under the coordination of internal support.

- ✓ **Key staff delayed: How will you work around their absence?**

Coordinate remote assistant with the main staff at the Chicago site.

8 **Timeline and Milestones:**

Below is a table detailing recovery timeline showing when we expect each critical system to be restored. Including key milestones and decision points.

| Critical System | Estimated restoration time | Key Milestone | Decision Points |
|---|---|---|---|
| Core Banking System | 6 hours | Estimated time adjusted after assessment, extent of data loss to be assessed. | How reliable is the data recovered?<br>What are the next options? |
| Online Banking Portal | 8 Hours | Based on the core banking restoration and availability of critical stuff and vendor to do manual configs | How is the system affected by CBS?<br>Vendor engagement?<br>Key staff engagement?<br>User credential affected? |
| Mobile Banking App | 8 hours | Based on core banking restoration | How is the extent of the damage?<br>Is user login affected or compromised? |
| ATM Network | 12 hours | Once the core banking system has been restored, we focus on ATM | Has the switch redirection affected properly?<br>Are the ATMs fully functional? |
| Wire Transfer System | 10 hours | Based on core banking and teller banking system | How many transactions have been affected?<br>Stopping the wire transfers temporarily until next day due to approaching deadline? |

| | | | |
|---|---|---|---|
| Loan Origination System | 48 hours | Based on all other system restoration this will be the last to be restored | Have all loans updated correctly? |
| Email System | 10 hours | We first concentrate on core banking | Have all communication restored? |
| Branch Teller System | 8 hours | Having all the tellers back online | Have all the systems restored?<br>How many tellers have been affected?<br>How many tellers are needed to function in a branch? |

**Task 2.2: Stakeholder Management**

Managed the various stakeholder concerns:

9 **Executive Communication:**

TO: CEO, CFO, COO, General Counsel

Subject: Update: System Outage-Primary Data Center Flooding

Executive Summary:

We are proving an update on current restoration activities, based on our assessment we have estimated the recovery time for our core banking restoration to take longer than our earlier estimated time, it will now take up 6 hours. We have also encountered challenges with Online banking portal which requires manual configuration that wasn't fully documented, we are contacting our vendor partner to assist with restoration, 3 key technical staff are also stuck in traffic due to weather and won't arrive for 2 more hours, which has limited our resources we have however gotten back up from our main site and secured remote assistance. We are currently focused on restoring CBS, mobile and online banking to mitigate customer complaints and reduce call to call center. We will need approval to engage with external experts on data restoration and also to engage our online banking vendor.

We will provide the next update at 10:00 am or sooner once we execute the current actions.

10 **Regulatory Communication:**

To: Regulatory Authority

Subject: Service Interruption Notification

Service outage affecting our primary data center

Pursuant of regulatory requirement, Finance First Bank is notifying you of an interruption in their services due to flooding incident that occurred as a result of a water main break on street outside the building hosting our data center. This incident has been classified as critical, and has affected our service including branch transactions, mobile and internet banking.

Details of the Incident: Tuesday early morning flooding that affected all services potentially affecting 25000 clients. We estimate the timeline for full restoration of the primary site to be 5-7 days, however current business continuity and disaster recovery has been activated and restoration services are underway with partial restoration estimated to start after 6 hours and full restoration to ensure normal business operation estimated at 24 hours.

Contact information: for questions regarding this notification, please contact Full Name, DR Mager, +1(xxx)-xxxxxxx, drmanager@financefirst.com

Sincerely,

Finance First Bank

11 **Customer Communication:** Draft a customer-facing statement for:

- ✓ **Website banner notification**

Temporary outage: systems unavailable, issue is being resolved, we regret the inconvenience. Update in 2 hours.

- ✓ **Social media post**

Dear customer, we are currently experiencing a temporary outage affecting our services, our team is working to resolve the issue as soon as possible. We will notify you once systems have been restored.

- ✓ **Branch signage**

Update: temporary service interruption affecting all our service, temporary pause on all wire transfers until further notice.

- ✓ **Call center talking points**

We are experiencing service outage temporarily affecting all our transaction, our team is working on restoring services, we regret the inconvenience and will update you once service has been restored.

12 **Media Management:**

First Finance Bank Responds to Service Outage

Chicago, Chicago – 09/December/2025– Finance First announced today that it has discovered and is actively responding to a system outage affecting all services.

Current Status: we experience a flooding incident that affected our primary data center which resulted to service outages.

Our Response: - We have contained the incident and taken steps to restore services from our recovery data center and are actively working towards full service restoration. We are notifying all affected individuals and regulatory authorities as required by law.

Commitment to Our Customers: our service availability and integrity of our system is our highest priority. We are committed to transparency and will provide updates as our system restoration continues.

Resources Available: Customers with questions can visit financefirstbank.com or call +1 (xxx)-xxxxxxx

About Finance First bank: Finance First Bank is a regional bank serving 250,000 customers across five states (Illinois, Indiana, Wisconsin, Iowa, Michigan). The bank operates 85 branch locations, , and offers retail banking, commercial lending, wealth management, and mortgage services.

Media Contact: Media Person, Media@financefirst.com , +1(xxx)-xxxxxxx

13 **Internal Communication:**

To: All Staff

Subject: Urgent: Service Outage

We are experiencing service interruption due to flooding that has affected our primary data center, affecting all systems, mobile, internet banking and wire transfers. Please handle appropriately with walk in clients and be on standby for official communication from communication teams. We will guide you in the next step.

We regret the inconvenience as our team works on service restoration. We will also notify you on possible time for recovery once we asses extent of damage.

## Phase 3: Complications and Decisions (Tuesday, 2:00 PM)

**SITUATION UPDATE - 2:00 PM**

### Our Response - Phase 3 (Due: Within 6 hours of Phase 1)

**Task 3.1: Critical Decision-Making**

14 **Data Integrity Decision:** We have chosen to pursue option B as it is the most viable option at this point. We provide detailed justification for our decision in the table outline below:

| Data Integrity | Business Impact | Risk Assessment | Customer Impact | Regulatory Implications | Resource Requirement |
|---|---|---|---|---|---|
| **Option A:** Use the 11:45 PM backup (clean data but 6.5 hours of lost transactions) | Possible financial impact in losses to either client (credits) or the bank (debit)<br>Reputational damage in customer incorrect statement | High, based on availability vs correctness | Inconsistent record, unbalanced statement<br>Loss of trust | AML non-compliance for the period not tracked<br>FDIC insurance and reporting requirements<br>Federal reserve oversight: affected due to incorrect position of reserves and money position overnight | Immediate: limited to service restoration<br>Long term: finance and reconciliation staff and technology to cross check the difference in the clean data and corrupt data. |
| **Option B:** Use the 6:15 AM backup and manually correct the 2,000 affected accounts (estimated 8-12 hours of work): **This is the safest option** | Resource intensive, possible reputational damage on the 2000 affected data | Medium, based on availability vs correctness, and also due to 2000 vs 25,000 clients | Minimal, small number of clients affected | AML non-compliance for the period not tracked<br>FDIC insurance and reporting requirements<br>Federal reserve oversight: affected due to incorrect position of reserves and money | Finace and Recons resources for manual reconciliation |

| | | | | | |
|---|---|---|---|---|---|
| | | | | position overnight | |
| **Option C:** Delay core banking restoration while investigating the corruption (unknown time) | Critical, possible panic and reputational damage, delayed restoration and continued down time | Critical based on system and service unavailability and the difference in RTO expectations | System unavailability, | State Banking regulation | Resource intensive in terms of cost and staff |

15 **Wire Transfer Priority Decision:**

How will you handle the wire transfer deadline? Options include:

- ✓ **Divert resources from core banking to prioritize wire transfers**

  This will derail core banking restoration activities and considering all systems ultimately rely on the CBS for correctness, this is not a viable option.

- ✓ **Accept missing the deadline and focus on core banking**

  This also is not optional, because the repercussion means accepting Significant customer dissatisfaction, Potential breach of contract penalties, Regulatory scrutiny, Reputational damage

- ✓ **Attempt parallel recovery of both systems**

  This is a viable option, which will allow for parallel restoration giving maximum effort to restoring both the CBS and wire transfer.

- ✓ **Seek alternative wire transfer processing method**

  This is viable only as mitigation once restoring the wire transfer system is unattainable one hour towards the deadline, in the meantime we have the alternative on standby.

**16 Justify your decision with risk-benefit analysis.**

In choosing the way forward, we looked at the potential benefit of having the wire transfer back up at the expense of the CBS.

17 **Vendor Licensing Issue:** Develop a plan to resolve the vendor licensing issue.

**What specific actions will you take**?

I will call for continuing with restoration services as we review our service level agreement and our escalation matrix,

**who will you escalate to**?

With the information on our escalation matrix, I will request our head of ICT to escalate on the vendor side and confirm continuity.

**What is your backup plan if the vendor doesn't respond?**

I will be looking at creating a backup from the current back up which will give us another 24 hours, or find a way to use the primary site license key which is currently not in use.

18 **Staff Management:**

**How will you address team fatigue while maintaining recovery momentum?**

Create a staffing plan for the next 12 hours including shift rotations, breaks, and backup resource deployment.

We have 25 staff actively from ICT all working either remotely or physically and other key point staff from critical business units, I will create a rotation where we ensure the staff are equally distributed with the key staff experienced on restoration rotated to cover each shift and divided equally such that each shift has a minimum of two experienced staff.

**Task 3.2: Risk Management and Contingency Planning**

19 **Risk Register:** below is a risk register, identifying the top 5 risks to successful recovery at this point in the scenario. For each risk, document we have documented as below:

| Risk Description | Probability | Impact | Mitigation Action | Contingency plan if risk materializes |
|---|---|---|---|---|
| Corrupt Data | High | High | Regular back up | Manual Reconciliation |
| Undocumented key steps | High | High | Keep technical specification documents | Vendor support |
| Core Banking License | High | High | License confirmation | Vendor engagement and support |
| Customer Panic | Medium | High | Regular notification | Frequent updates and individual communication |
| Resource Fatigue | Medium | High | Staff rotation | Forced rotation and training |

20 **Go/No-Go Decision:**

**At what point would you decide to abandon the DR site recovery and pursue an alternative approach?**

If all restoration efforts have failed to materialize, core banking license has not been acquired and wire transfer restoration fails.

**Define your decision criteria and alternative options.**

The alternative option will be to announce temporary system unavailability.

# Phase 4: Final Recovery and Transition (Wednesday, 8:00 AM)

**SITUATION UPDATE - Wednesday 8:00 AM (26 hours after incident)**

## Our Response - Phase 4 (Due: Within 8 hours of Phase 3)

**Task 4.1: Transition Planning**

21 **Return Strategy:** Develop a strategy for transitioning back to normal operations. Should you:

- ✓ **Return to the Chicago primary site once restored?**

This is the most viable option based on the cost of operating the DR site per day and also staff challenges with logistics, at the same time caters for business continuity.

- ✓ **Continue at DR site indefinitely?**

This will be cost intensive and also potential loss of resources (staff) who have already raised concerns on the logistics.

- ✓ **Implement a hybrid approach?**

This will likely result in system outages and might lead customer inconvenience leading to panic and loss of trust.

22 **Justify your recommendation with cost-benefit analysis and risk assessment.**

The most viable option at this stage is to fully restore the main site, confirm it is fully restored, communicate to clients on system migration prior to moving back to the primary site with explained downtime estimates, preferably over the weekend where there are no deadlines and walk-in clients.

23 **Transition Timeline:** below is a detailed timeline for the chosen transition approach:

| Step | Key Milestones & activities | Testing and Validation Requirements | Cutover Procedures | Rollback Plan if transition fails |
|---|---|---|---|---|
| Days 1-2 | Assess the extent of damage, consult insurance | Test all infrastructure and systems | Insurance report complete | N/A |
| Day 3 | Consultation with insurance and equipment restoration | Confirm how long it will take to replace damaged assets | Invoice payment and asset ordering | N/A |
| Day 4-5 | Rebuilding the DR site, the HVAC system and the server racks | Work with vendor support to test how ready the site is | Complete test results on the site | N/A |
| Day 6-7 | Data restoration and testing, communication to stakeholders and clients on migration and possible temporary intentional, voluntary downtime | Confirmation from subject matter experts that the systems and data are correct.<br>Communication completed | Data confirmation and stakeholder communication | Pause on migration based on feedback and approval following recent outages. |
| Day 8-9 | Migration preparation | Staff rotation and availability confirmation, | Approvals from Exco and Board,<br>Complete staff schedule and | N/A |

| | | | | |
|---|---|---|---|---|
| | | System back up confirmation<br>Data correctness confirmation | resource confirmation,<br>Confirmed data backup procedures | |
| Day 10 | Migration | Migration from DR site to Primary site | All data is correct, all systems restored, all client transactions processing, | Restore back to the data recovery center. |

24 **Business Resumption:** What steps will you take to fully resume normal business operations? Address:

- ✓ **Customer notification and confidence rebuilding**

  Send out notification to all clients that all systems have been restored, appreciate the patience and as a gesture of goodwill mobilize customer resumptions with acceptable discounts.

- ✓ **Staff return to normal schedules**

  Allow for the staff who were stand-by on recovery to go on vacation and recreate the schedules to reflect current availability and a phased approach towards long-term rotation. Also focus on training to ensure all staff are prepared for recovery to avoid fatigue.

- ✓ **Vendor and partner communications**

  Communicate to the vendors with clear action points and a full update on the current status.

- ✓ **Regulatory close-out**

  Formally write to the regulators on the action points and confirm availability for a meeting to discuss, what went well, what went write and opportunities for improvement as well confirmation of business resilience.

## Final Deliverable: Comprehensive After-Action Report

**Task 5.1: After-Action Report (AAR)**

This section has been documented in AAR document also attached as part of this assignment submission.